Objective

To implement and demonstrate NegativeArraySizeException in Java using try-catch by creating an array with a negative size.

Steps for Execution

1.Create a Java project and package named javacore.

2.Create a file named NegativeArrayExample.java and enter the given code.

3.Declare the array size as -5.

4.Try to create an array using the negative size.

5.Handle the exception using catch.

6.Compile and run the program.

7.Observe the output.

Algorithm

1.Start the program.

2.Set the array size to -5.

3.Try to create an integer array with the given size.

4.Catch NegativeArraySizeException.

5.Display the error message.

6.Display "Program ended.".

7.Stop the program.

UML Class Diagram

```
┌──────────────────────────────┐
│   NegativeArrayExample   │
├──────────────────────────────┤
│ + main(args : String[])  │
│         : void         │
└──────────────┬───────────────┘
```

```
          │
          │ creates
          ▼
┌─────────────────────────────┐
│       int[] Array           │
├─────────────────────────────┤
│ Size = -5                   │
└─────────────┬───────────────┘
          │
          │ throws
          ▼
┌─────────────────────────────┐
│  NegativeArraySizeException │
├─────────────────────────────┤
│ + Exception handling        │
└─────────────────────────────┘
```

Relationship

- NegativeArrayExample attempts to create an integer array.
- A negative array size (-5) causes a NegativeArraySizeException.
- The exception is handled using the catch block.

CODE:

```
package javacore;

public class NegativeArrayExample {

    public static void main(String[] args) {


        int size = -5;
```

```
        try {
            System.out.println("Array size: " + size);

            int[] numbers = new int[size];

            System.out.println("Array created successfully.");
        }
        catch (NegativeArraySizeException e) {
            System.out.println("Error: Array size cannot be negative.");
        }

        System.out.println("Program ended.");
    }
}
```

Output

Array size: -5

Error: Array size cannot be negative.

Program ended.

Result

The program successfully demonstrates NegativeArraySizeException handling in Java. Since an array cannot be created with a negative size, NegativeArraySizeException occurs and is handled using the catch block.